ETHGlobal showcase intelligence for mindX expansion

The ETHGlobal showcase contains roughly **80 strategically relevant projects with public GitHub repositories** that map cleanly onto mindX/PYTHAI/DELTAVERSE components, concentrated heavily in 2025–2026 hackathons (HackMoney, ETHGlobal New Delhi, Cannes, New York, Buenos Aires). The most consequential finding: **ERC-8004 has crossed from draft EIP (Aug 2025) into working hackathon implementations**, and a small cluster of projects — Hubble Trading Arena, HyperAgent, AgentPass, Router402, The42 — now combine ERC-8004 identity with x402 payment rails in production-leaning architectures. **These are the closest existing analogs to the mindX vision and should be studied first.** Conversely, ETHGlobal is a near-empty well for Algorand-native code (only two relevant projects, both from the Unite DeFi 1inch hackathon) and contains zero BitVM/sBTC/Babylon/Citrea work — those ecosystems will require off-platform sourcing. The remainder of this report inventories projects by capability area, flags cross-cutting candidates, and concludes with a ranked top-15 integration roadmap.

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1. Agent identity, reputation, and ERC-8004 implementations

This is the most strategically dense category, mapping directly onto **BONAFIDE's Genius/Fides/Tessera contracts** and **BANKON's AlgoIDNFT**.

HyperAgent (https://ethglobal.com/showcase/hyperagent-mzvri) is **the single closest architectural analog to mindX found anywhere on ETHGlobal**. It implements ERC-8004 registries on Hedera with a **HYPT token lock-and-burn registration**, **TEE-based verifier agents** (Oasis ROFL), Google A2A peer protocol, and Foundry-based modular contracts. The HYPT lock-and-burn pattern maps directly onto DELTAVERSE token-gated agent registration; the TEE verifiers map onto openBDK's 1+3 BFT validators; the ERC-8004 registries map onto BONAFIDE's Genius (identity)/Fides (trust)/Tessera (badges) trio. **This project should be the first one cloned and read end-to-end.** ETHGlobal

RoboAdvisor ERC-8004 (https://ethglobal.com/showcase/roboadvisor-erc-8004-nr8uz, ETHGlobal New Delhi 2025) is the canonical reference implementation of the three-registry pattern (Identity NFT + Reputation + Validation) on **Rootstock** — useful because it co-locates ERC-8004 with Bitcoin sidechain logic, doubly relevant to DELTAVERSE.

AgentPass (https://ethglobal.com/showcase/agentpass-92p9u, GitHub: github.com/kai-agent-free/AgentPass) is **production-leaning code for the agent-identity-plus-spending-policy primitive**: Ed25519 cryptographic passport per agent, World ID human-backing, ENS naming, Ledger hardware backing, MCP protocol, and a published `@agentpass/sdk` with **320 tests**. This is the cleanest BANKON-side template available — local-only private keys with an on-chain public-key registry plus challenge-response auth and a built-in trust score that scaffolds BonaToken. GitHub + 5

Hedera Agent Trust / Zkred Agent ID (https://ethglobal.com/showcase/hedera-agent-trust-mur21) ships as a reusable **npm plugin** for the Hedera Agent Kit. Its DID-signature-based agent-to-agent authentication is the pattern AlgoIDNFT soulbound DIDs need to integrate. ETHGlobal ETHGlobal

Reputable.Network (https://ethglobal.com/showcase/reputable-network-2hxgn) is **the strongest BONAFIDE analog**: hybrid on-chain/off-chain reputation with **tradable ERC-721 reputation models**, Sign Protocol attestations, Blockscout indexing, and Web3Auth GitHub-linking. It maps onto BonaToken (the token), Tabularium (records), Fides (scoring), and Senatus (governance of model parameters) simultaneously. ETHGlobal

Signet (https://ethglobal.com/showcase/signet-v949x) deserves special attention: World ID-sealed agents whose **autonomy level, risk profile, and spending ceilings are stored as ENS subname text records** under signet.eth. This is perhaps the most BONAFIDE-aligned single project — Genius (sealed identity) + Censura (autonomy levels) + SponsioPactum (spending commitments) in one package.

Other notable identity/reputation entries: **Nomis** (cross-chain wallet scoring across 20+ L1/L2s), **Trust Explorer** (EAS + WorldID custom-resolver pattern), **Honorus** (ENS+Filecoin+Lit), **Post Quantum Agent** (ML-DSA-44 hybrid signing, forward-looking BANKON hardening), **AgenticPlaceholder** (notable name overlap with mindX's AgenticPlace; uses True Network attestations and Coinbase AgentKit dual agents — worth direct review). ETHGlobal + 6

2. Soulbound tokens, DIDs, and W3C verifiable credentials

Soulmates (https://ethglobal.com/showcase/soulmates-wveqg, github.com/rohanagarwal/hack-fevm-submission) is the most complete **SBT issuance engine** on the showcase: 17 contracts and 11 test files implementing a Core/SBT/Factory/Oracles/Storage architecture. It's the cleanest template for BONAFIDE's Genius (issuance) and Tessera (badges) contracts; the factory pattern enables per-issuer credential schemas without redeploying core logic. ETHGlobal ethglobal

IPDID (https://ethglobal.com/showcase/ipdid-speed-up-did-resolution-h5dd3, github.com/IPDID/project-README) is **the closest end-to-end W3C DID stack to BANKON's AlgoIDNFT**: Expo React Native DID Wallet (iOS+Android), Node CLI with libp2p+IPFS+IPLD+CID, SKALE smart contracts via Hardhat, DIF Universal Resolver Driver (Docker), QR-code VC issuance. Multi-component framework, not demo glue. ethglobal ETHGlobal

Iris-Bound Tokens (https://ethglobal.com/showcase/iris-bound-tokens-3neee, ETHBogotá Worldcoin Best Overall) provides **the design rationale for binding identity to a person rather than a wallet** — critical for AlgoIDNFT recoverability if a key is lost. The contracts explicitly reject ERC-1238/ERC-4973 in favor of ERC-721 with disabled transfers plus Worldcoin nullifier-hash mapping. ETHGlobal ethglobal

Attestation Explorer (https://ethglobal.com/showcase/attestation-explorer-4b5o6) is **uniquely valuable as a converter library between W3C VCs and EAS schemas**. Because BANKON speaks W3C DID/VC and BONAFIDE Tabularium speaks EAS-style attestations, this code is the bridge artifact between the two systems.

Token Bound Accounts (https://ethglobal.com/showcase/token-bound-accounts-kg5gr) provides a clean **ERC-6551** reference — letting each Genius identity NFT own its own programmable account (treasury, history). Combined with Soulmates' factory pattern, this gives BONAFIDE Genius a complete primitive set. ETHGlobal

Sign Protocol cluster — **Clarity** (Schema Hooks for custom attestation logic), **Eth GovOps** (full enterprise governance/work-order suite mapping to Senatus + SponsioPactum), **CryptoGuard** (off-chain gasless attestations — scarce and important for high-volume agent activity), **Sanicert** (ZK age/attribute proofs via circom + EdDSA) — collectively constitute a reusable attestation primitive library for Tabularium and Tessera.

EAS cluster — **AttestDeFi** (token distribution gated by attestations), **VouchSafe** (revocable WorldID-attested reputation), **PassportGlobal** (Foundry-based EAS stamps — rare Foundry hit), **OSS on OP Directory** (schema design pattern for artifact↔identity mapping). ETHGlobal + 2

Mango (https://ethglobal.com/showcase/mango-azfbe) demonstrates **Sismo-gated Safe Module ownership updates** — the cleanest pattern for Senatus governance where Safe ownership requires ZK proof of credential possession. ETHGlobal

3. Autonomous agent infrastructure and LLM tooling

AgentNet (https://ethglobal.com/showcase/agentnet-c8qzg) implements **libp2p DHT-based peer discovery + Noise Protocol encryption + Coinbase CDP AgentKit + GPT-4o** for fully on-chain agent transactions. The libp2p discovery layer is reusable for openBDK validator coordination; the multi-agent constellation pattern (User Agent → specialized agents) maps onto AgenticPlace's Soul-Mind-Hands cognition. ETHGlobal ETHGlobal

AgentNexus (https://ethglobal.com/showcase/agentnexus-5jcq5, github.com/irajgill/AgentNexus) deserves a deep read: **Fetch.ai uAgents + ASI:One LLM + MeTTa knowledge graphs + on-chain Groth16 StrategyVerifier (Circom + snarkjs) + Avail Nexus bridgeAndExecute() + Foundry**. The ZK strategy commitment pattern is reusable for openBDK validator-vote privacy; MeTTa knowledge graphs are an explicit Mind-layer template. ETHGlobal + 3

E-Lisa Simpson (https://ethglobal.com/showcase/e-lisa-simpson-n7g5q) is the ElizaOS fork that integrates **EIP-7702 + ERC-7579 + ERC-7715 (Smart Sessions)** with Privy Server Wallets and Safe smart accounts. This is the canonical pattern for an agent that operates on a user wallet **without ever touching the private keys** — the AgenticPlace Hands layer needs exactly this safety primitive. ETHGlobal

AgentForge (https://ethglobal.com/showcase/agentforge-h0047) uses **Lit Protocol's Agent Wallet** (MPC keys with policy-based execution) plus ERC-4626 vaults plus Chainlink CCIP. The Lit MPC + policy-bound key pattern is the de-facto safe pattern for autonomous agents and should inform mindX agent custody. ETHGlobal + 2

HyperMCP (https://ethglobal.com/showcase/hypermcp-5m0vj) provides a **production-quality MCP server** integrating Envio's HyperIndex/HyperSync/HyperRPC across 70+ EVM networks — 35+ tools exposed to Claude Desktop via Model Context Protocol. Reusable as the indexing-MCP backend for AgenticPlace. ETHGlobal

ChainGPT (https://ethglobal.com/showcase/chaingpt-and-accountability-for-ai-agents-doquk) is conceptually critical: **C++ deterministic LLM inference run across multiple validator nodes with on-chain dispute mediation**. This is the BFT-applied-to-LLM-output pattern that openBDK could extend to validate agent reasoning, not just transactions. ETHGlobal

ZEECA uses zkTLS + Phala Network TEEs + LangChain to bring Web2 compliance into Web3 privately; the TEE-based private execution pattern is useful for openBDK secure enclaves. ETHGlobal

AGI Solver (https://ethglobal.com/showcase/agi-solver-7pzu8) implements a **modular state machine for agent intent execution** — the separation of agent intent (publish to contract) from execution (solver) is exactly the AgenticPlace Soul-Mind-Hands separation, with LiFi DEX integration as a working example. ETHGlobal

Other strong entries: **Power Agents** (XMTP group chats with per-agent ENS subdomain + CDP wallet + recurring-tx scheduler — directly relevant for spawning child agents in AgenticPlace), **Etherius** (uAgents + ASI:One + OpenSea MCP + x402 — best Fetch.ai-plus-x402 template), **Stableburn x402** (agent that earns x402 revenue and burns tokens — relevant for THRUST/PAIMINT/PAI tokenomics), **TradeAlgoAVS** (EigenLayer AVS + Autonome + Nillion SecretVault + on-chain operator reputation — full BONAFIDE+DELTAVERSE+openBDK reference architecture). GitHub + 10

4. x402 payment rails and agent-to-agent micropayments

Important Algorand finding upfront: no ETHGlobal showcase project implements x402 on Algorand. All x402 ETHGlobal projects target USDC on Base/Polygon/Arc/Sepolia. The patterns (EIP-3009 transferWithAuthorization, EIP-712 payment intent, facilitator/escrow contracts) are nonetheless directly portable as design references; the chain-specific Algorand bridge code will need to come from Algorand Foundation's own SDK/docs.

MCPay.fun (https://ethglobal.com/showcase/mcpay-fun-y16d3, github.com/microchipgnu/mcpay.fun, ETHGlobal Prague 2025 finalist) is **the cleanest x402 SDK on the showcase**: complete 402→pay→retry loop as a reusable TypeScript SDK plus CLI proxy

adapter plus MCP server. The "wrap any API in x402" model is exactly what BANKON needs to expose paywalled mindX agent resources. ETHGlobal ethglobal

Router402 (https://ethglobal.com/showcase/router402-b717q, github.com/itublockchain/hackmoney-router402, HackMoney 2026 finalist) is **production-leaning code**: Turborepo + Bun, **forked OpenFacilitator for Base Flashblocks (~0.2s settlement)**, Pimlico + ZeroDev smart accounts with session keys, gasless via paymaster, and a published `@router402/sdk`. The session-key pattern (no popups, gasless) is the model for mindX agents that must pay autonomously without per-call human approvals. GitHub + 4

The42 (https://ethglobal.com/showcase/the42-sijqk, github.com/ekailabs/ethglobal-nyc, ETHGlobal NYC 2025 — Best AI App on Dynamic + Best AI App on Fern) implements **x402 streaming in PYUSD with EIP-7702 policy-gated smart accounts** enforcing per-call caps, daily limits, and allowed recipients via session keys. **Critically, it builds a custom x402 facilitator for non-USDC stablecoin (PYUSD) — exactly the template if DELTAVERSE-issued tokens (THRUST/PAIMINT/PAI) need to settle x402 calls.** ETHGlobal + 3

x402-flash (https://ethglobal.com/showcase/x402-flash-yj9wd, ETHGlobal NYC 2025 Coinbase CDP Track winner) is **Foundry-tested + Nora-audited escrow**. It introduces a "FLASH" x402 scheme where pre-funded escrow lets servers respond instantly without waiting for on-chain settlement — directly applicable to BANKON streaming-pay model and high-frequency agent API calls where 200ms latency is unacceptable. Contract pattern is **chain-agnostic and re-implementable in PyTeal for Algorand**. ETHGlobal ETHGlobal

Private-Escrow x402 (https://ethglobal.com/showcase/private-escrow-x402-fj0v1, github.com/hamiha70/x402-escrow) implements three progressive schemes — exact (sync), escrow-deferred (instant + batched), private-escrow (ZK planned via Noir/Aztec). The **batched settlement pattern (~3k gas/payment when amortized over 100 payments)** is the only economically viable approach for mindX agents making thousands of cheap calls. Multi-chain Foundry implementation across Base/ETH/Arbitrum/Optimism Sepolia, Polygon Amoy, Arc Testnet. ETHGlobal + 6

Hubble Trading Arena (https://ethglobal.com/showcase/hubble-trading-arena-oddx1, ETHGlobal Buenos Aires 2025 top-10 finalist out of 475 projects) is **the most complete published x402 + ERC-8004 multi-agent fleet architecture** — thousands of autonomous financial agents collaborating on-chain. ERC-8004 gives portable agent identity; x402 gives the payment rail. **This is the closest published architecture match for the DELTAVERSE multi-agent payment vision and validates feasibility at scale.** X

Synapse (https://ethglobal.com/showcase/synapse-8m2sw, github.com/Synapse-Labs-X/synapse-protocol) is explicitly framed as a **payment-layer protocol for AI agents** with USDC microtransactions and multi-agent task routing — Dockerized and protocol-shaped rather than demo-shaped. ETHGlobal + 2

0pi (https://ethglobal.com/showcase/0pi-uybzg, github.com/dizzydes/0pi) is uniquely valuable because it's **Python FastAPI** (rare on ETHGlobal — most x402 work is TypeScript), making it directly compatible with mindX's Python stack. Wraps any existing API in an x402 response. ETHGlobal + 3

Paybot (https://ethglobal.com/showcase/paybot-q7grd, github.com/superposition/paybot, ETHGlobal Buenos Aires 2025 top-10) is **the most polished x402 facilitator implementation** in the showcase: Solidity QUSDToken (ERC20 + EIP-2612 permit) + Escrow contract on Base, TypeScript+Hono x402 facilitator, Hardhat, Docker, demonstrating gasless dual EIP-712 signatures. ETHGlobal +2 ethglobal

Payload Exchange (https://ethglobal.com/showcase/payloadexchange-x07pi, ETHGlobal Buenos Aires 2025 top-10) implements **payment-method-abstraction over x402** — sender chooses any token, receiver gets stable, sponsor pays partial/full in exchange for actions/data. Useful if DELTAVERSE token holders should pay for resources priced in USDC. WEEX

zkx402 / ProofofLeak (https://ethglobal.com/showcase/zkx402-proofofleak-wxtjf, ETHGlobal Buenos Aires 2025 top-10) extends x402 with ZK proofs for identity-tier pricing (human vs AI-agent) and content-provenance proofs. WEEX ODaily

SubPay AI uses Flow Actions wrappers (atomic, agent-friendly Actions) for subscription pipelines on Flow EVM with Foundry tests, while **BlinkStream** demonstrates **state-channel session vaults** for batched stablecoin micro-payments — alternative architectural model to escrow-x402 if mindX agents have predictable session-bounded budgets. ETHGlobal ETHGlobal

5. Cross-chain infrastructure and BFT validator networks

This category contains the strongest matches for **openBDK (1+3 BFT consensus) and DELTAVERSE multi-chain coordination**.

Nexus Network (https://ethglobal.com/showcase/nexus-network-ueajk, ETHOnline 2023, github.com/Nexus-2023) is the **closest production analog to openBDK on the showcase**: it modifies the **Polygon zkEVM LxLy bridge** to stake locked ETH via SSV Network DVT (Distributed Validator Technology). DVT is essentially BFT distributed signing across multiple validator nodes — directly maps to openBDK's 1+3 (Relayer + 3 token-gated Validators with 2/3 quorum). Pluggable contracts forked across zkEVM/Scroll/Mantle. ETHGlobal ETHGlobal

ETL Network (https://ethglobal.com/showcase/etl-network-6b6a8) is **the design twin of openBDK**: a Kyve-blockchain consensus pool of nodes executing the same JS function and collecting Gnosis-Safe multisig signatures (votes), then producing a single Polygon transaction once threshold is met. Replace Kyve with token-gated validators and 2/3 quorum and you have openBDK. **This is the project whose architecture diagram should be copied first.** ETHGlobal

CrossSync Protocol (https://ethglobal.com/showcase/crosssync-protocol-da47b) provides a **chain-agnostic multi-bridge gateway** unifying LayerZero, Hyperlane, Wormhole, Axelar, CCIP,

and Connext under a single `sendMessage`/`receiveMessage` interface, deployed on Polygon zkEVM, Mantle, Scroll, and others. This is the transport layer abstraction openBDK + DELTAVERSE need: validators can vote on which underlying transport to use without coupling to one provider. ETHGlobal

Whisper (https://ethglobal.com/showcase/whisper-j8j2w, github.com/Zhixuan0318/Whisper-ETHGlobal, ETHGlobal NYC 2025 LayerZero 3rd place) is a **no-code LayerZero V2 channel deployer explicitly for omnichain AI agents**. The webhook-with-HMAC verification pattern can be reused as openBDK's Relayer-to-Validator authenticated channel; the listed "Omnichain DAO" use case exactly matches mindX/AgenticPlace cross-chain agent coordination. ethglobal + 4

HyperWallet (https://ethglobal.com/showcase/hyperwallet-ye9zz, ETHGlobal Singapore 2024 Hyperlane winner) implements **Hyperlane Interchain Accounts** as a chain-abstracted wallet — single EOA controls many chain wallets via CREATE2 address computation. Reusable as a library (published codegen + actions packages). Directly applicable to mindX agent identity across DELTAVERSE chains. ethglobal + 2

Cross Chain Safe (https://ethglobal.com/showcase/cross-chain-safe-f5w5g, ETHGlobal Istanbul Safe 3rd place, github.com/cross-chain-safe/contracts) implements Safe-on-A controls Safe-on-B via Hyperlane — precisely the BONAFIDE Senatus → per-chain Censura execution pattern. ETHGlobal ethglobal

Omnichain Governance / OmniGov (https://ethglobal.com/showcase/omnichain-governance-gmfuy, github.com/markcarey/omnigov) is **production-ready BONAFIDE Senatus**: OpenZeppelin Governor with overridden execute, LayerZero ILayerZeroReceiver, ReceiverExecutor pattern with allowlist (only Governor on governance chain). Tally UI compatible. ethglobal ETHGlobal

OmniVote (https://ethglobal.com/showcase/omnivote-4r0jb, github.com/janvinsha/omnivote, ETHOnline 2024 Sign Protocol winner) implements primary-chain governance state with secondary-chain vote relaying via Chainlink CCIP plus on-chain attestation of vote integrity. ETHGlobal ethglobal

InfiniteDAO (https://ethglobal.com/showcase/infinitedao-inqai, ETHGlobal Bangkok 2024) mirrors openBDK structurally: an Orchestrator contract listens to CCIP receiver state changes — Orchestrator = Relayer; CCIP receivers = Validators reporting in. ethglobal ETHGlobal

FlashFi (https://ethglobal.com/showcase/flashfi-g27p4, ETHGlobal Bangkok 2024, **4-prize winner including LayerZero 2nd place**) combines LayerZero OApp+OFT cross-chain messaging with an XMTP-based AI agent for multisig wallet creation — direct template for AgenticPlace DAIO with cross-chain treasury operations. ethglobal ETHGlobal

Unify (https://ethglobal.com/showcase/unify-make-your-aa-cross-chain-a4x7h) is one of the few projects using the **Polygon zkEVM native LxLy bridge** (predecessor to AggLayer Unified Bridge), with Gnosis Safe modules for cross-chain account abstraction. ETHGlobal ETHGlobal

Fluvia (https://ethglobal.com/showcase/fluvia-4wkgv) demonstrates **CREATE2 same-address contract deployment across chains plus Circle CCTP** — critical for DELTAVERSE Debt Inheritance unique-identity-across-chains. ETHGlobal

Notable gap: no AggLayer Unified Bridge winner with public GitHub surfaced in searches. ETHGlobal Brussels offered an AggLayer bounty but specific winning entries were not retrievable. Recommend checking github.com/agglayer/agglayer-contracts directly for Polygon Labs's canonical implementation.

6. Algorand integrations and Bitcoin DeFi

ETHGlobal is overwhelmingly Ethereum-centric. **Only two substantive Algorand projects exist on the showcase**, both from Unite DeFi (1inch hackathon, 2025):

algobridge (https://ethglobal.com/showcase/algobridge-0i64b, github.com/Ashar20/fusion-cross-chain-bridge) is **the highest-fit project for Parsec x402 Algorand rails on the entire showcase**: gasless ETH↔ALGO swaps via 1inch Fusion+ architecture, **PyTeal Algorand HTLCs** (closest to Algopy stack), official 1inch Escrow Factory, and a multi-resolver relayer system with four strategies (high-frequency, arbitrage, MEV, conservative) using Dutch-auction bidding. Library-grade with EIP-712 intent signatures, partial-fill support, and clean separation of `.env.relayer` / `.env.resolvers`. ethglobal + 5

AlgoSwap (https://ethglobal.com/showcase/algoswap-b0aw0, github.com/Potti1234/PottiSwap, **1st place 1inch "Extend Fusion+ to Any Other Chain"**) uses the **Algorand TypeScript SDK** rather than PyTeal but provides the cleanest demonstration of an EVM↔Algorand HTLC swap with relayer, secret reveal, and escrow on both sides. ETHGlobal ethglobal

For BANKON AlgoIDNFT specifically, ETHGlobal is a dead end — no W3C DID work on Algorand exists on the showcase. The lead researcher should look at Algorand Foundation's own hackathons, the `awesome-algorand` GitHub topic, AlgoKit Utils repos, and ARC-72/19 NFT standard reference implementations directly.

For Bitcoin DeFi, the highest-value findings are:

StarBTC (https://ethglobal.com/showcase/starbtc-vsaev, github.com/songsjun/StarBTC, StarkHack) is **the highest-fit project for DELTAVERSE Debt Inheritance Protocol**: native Bitcoin transaction script for BTC locking + EVM smart contract for loan terms + BEL2 arbiter network for arbitration + StarkWare Cairo 1 ZK-circuits for BTC transaction proofs. **No WBTC, no bridge — pure BTC mainnet with non-liquidatable loans.** This is the closest ETHGlobal analog of "BTC as hard-money anchor with EVM smart contract layer." ETHGlobal +2 ETHGlobal +2

TitanFi (https://ethglobal.com/showcase/titanfi-k1oj6, github.com/RahulBansal123/titan-finance, ETHOnline 2024) implements **self-repaying zero-interest loans on Rootstock with Foundry tests** — collateral generates Sovryn yield that auto-repays the user's loan. Directly

analogous to the "Debt Inheritance" auto-servicing concept and matches mindX's Foundry preference. ETHGlobal

ThunderPortal (https://ethglobal.com/showcase/thunderportal-haf30, github.com/NuttakitDW/thunder-portal, Unite DeFi) provides a **Rust Bitcoin HTLC service** with real script generation and testnet3 monitoring, plus a Solidity Factory pattern on Sepolia. Most production-ready BTC↔ETH atomic swap on the showcase. ethglobal + 2

OrdiSynth (Rootstock Best Runes/Ordinals winner at ETHOnline 2024) wraps Bitcoin-native Ordinals/Runes as ERC20 — relevant if mindX ever needs Bitcoin-native assets as DELTAVERSE collateral.

RootWallet demonstrates **Lightning Network ↔ RSK bridging via Voltage Lightning-as-a-Service** — useful if Parsec ever wants Lightning rails alongside Algorand x402. ETHGlobal

Confirmed gaps: no BitVM, sBTC/Stacks, Babylon, Citrea, or BOB (Build on Bitcoin) projects on ETHGlobal — those ecosystems run their own hackathons.

7. Cross-cutting projects spanning multiple capability areas

Several projects span three or more capability areas and deserve coordinated study:

Project	Identity	Infra	Payment	Cross-chain	Notes
HyperAgent	ERC-8004 + HYPT lock	TEE verifiers + A2A	HYPT token	—	Closest mindX architecture analog
Hubble Trading Arena	ERC-8004	Multi-agent fleet	x402	—	Most complete x402+ERC-8004 published
AgentNexus	uAgents identity	uAgents + ZK	Bridge-and-execute	Avail Nexus	Foundry, ZK strategy verifier
TradeAlgoAVS	Operator reputation	EigenLayer AVS	Strategy fees	—	Full BONAFIDE+DELTAVERSE+openBDI ref
FlashFi	Multisig agent	XMTP AI agent	OFT cross-chain	LayerZero	4-prize winner ETHGlobal Bangkok
Signet	World ID + ENS	Multi-agent Claude	Spending ceilings	—	BONAFIDE-aligned end-to-end
The42	EIP-7702 policy	LLM inference	x402 PYUSD	—	Non-USDC x402 facilitator template
Whisper	LayerZero OApp	Omnichain agents	—	LayerZero V2	Channel-as-a-service for AI agents
AgentForge	Agent identity	Lit MPC wallet	ERC-4626 yield	Chainlink CCIP	Policy-bound agent custody
Etherius	Multi-agent identity	Fetch.ai + MCP	x402 + CDP	—	Best uAgents+x402 combo

The **AgenticPlaceholder** name overlap with mindX's AgenticPlace is significant and warrants direct review by the lead architect to resolve any branding/conceptual collision before public

launch — that project uses Coinbase AgentKit dual agents, True Network attestations, and a PSDT token in a complete on-chain AI agent economy. ETHGlobal ETHGlobal

Strategic top-15 integration roadmap

The recommendations below are ranked by **strategic value × code reusability × architectural fit with mindX**, accounting for the constraint that hackathon code is rarely production-ready but often contains reusable patterns and contracts.

Tier S — clone, read end-to-end, fork patterns immediately:

1. **HyperAgent** — closest existing architectural analog to mindX. ERC-8004 + token-gated registration + TEE validators + A2A peer protocol. Single most important project to study.
2. **AgentPass** — production-quality SDK with 320 tests for agent identity + spending policy. Direct BANKON template combining hardware-bound identity, World ID, and ENS.
3. **Hubble Trading Arena** — proves x402 + ERC-8004 multi-agent fleet feasibility at scale. Validates the DELTAVERSE multi-agent payment vision.
4. **Router402** — production-leaning x402 SDK with session keys, smart accounts, gasless paymaster, and forked Flashblocks facilitator. Template for autonomous mindX agents paying per LLM call.
5. **The42** — only project with a custom x402 facilitator for non-USDC stablecoin (PYUSD), making it the direct template for DELTAVERSE-issued THRUST/PAIMINT/PAI x402 settlement.
6. **MCPay.fun** — cleanest x402 SDK + CLI proxy + MCP server combination. The "wrap any API in x402" pattern is exactly what BANKON needs.
7. **ETL Network** — design twin of openBDK: explicit Relayer + Validator + threshold-signing pattern on Polygon. Architecture diagram should be borrowed directly.
8. **StarBTC** — highest-fit reference for DELTAVERSE Debt Inheritance: native BTC + EVM contracts + arbiter network + ZK proofs, no wrapped tokens.
9. **algobridge** — highest-fit project for Parsec x402 Algorand rails. PyTeal HTLCs + multi-resolver gasless relayer system; the only credible Algorand template on ETHGlobal.

Tier A — strong patterns, integrate selectively:

10. **Reputable.Network** — strongest BONAFIDE analog with tradable ERC-721 reputation models, Sign Protocol attestations, and hybrid on/off-chain data sources. ETHGlobal
11. **Soulmates** — most complete SBT issuance engine for Genius/Tessera contracts: 17 contracts, 11 test files, factory pattern. ETHGlobal

12. **Nexus Network** — DVT validator network grafted onto Polygon zkEVM LxLy bridge; closest production analogue for openBDK at scale across Polygon CDK chains.
13. **TitanFi** — Foundry-tested self-repaying loans on Rootstock with Sovryn yield adapters; directly maps to DELTAVERSE auto-servicing debt. ETHGlobal
14. **Omnichain Governance / OmniGov** — production-ready BONAFIDE Senatus on OpenZeppelin Governor + LayerZero with Tally UI compatibility.
15. **x402-flash** — Foundry-tested + Nora-audited escrow contract solving the streaming-payment latency problem; pattern is chain-agnostic and re-implementable in PyTeal.

Tier B — consult for specific patterns: Whisper (LayerZero V2 omnichain agent channels), HyperWallet (Hyperlane ICA agent identity), CrossSync Protocol (multi-bridge gateway abstraction), E-Lisa Simpson (EIP-7702/7579/7715 Smart Sessions for keyless agents), AgentForge (Lit MPC policy-bound agent wallets), AgentNexus (multi-agent + ZK strategy verification + Foundry), IPDID (W3C DID end-to-end stack closest to AlgoIDNFT), Iris-Bound Tokens (person-bound vs wallet-bound identity rationale), Attestation Explorer (W3C VC ↔ EAS converter), Mango (Sismo-gated Safe Module for Senatus), Signet (World ID-sealed agent metadata in ENS), ThunderPortal (Rust BTC HTLC service), AGI Solver (intent-inbox pattern for AgenticPlace Hands), HyperMCP (production MCP server pattern).

Conclusion and key takeaways

The ETHGlobal showcase reveals a **late-2025-to-2026 inflection point** where ERC-8004, x402, and agent-frameworks coalesced into recognizable production-leaning architectures — and **mindX is well-positioned because its component decomposition (Soul-Mind-Hands, Genius/Fides/Tessera, Relayer+Validators) maps unusually cleanly onto the patterns now emerging at top-prize level on the showcase**. Three architectural insights matter most. First, **HyperAgent and Hubble Trading Arena have already validated the core mindX thesis** — token-gated agent registration plus ERC-8004 plus x402 plus TEE/AVS validation is a coherent, demonstrable stack. Second, **the cleanest reusable code lives in SDKs, facilitators, and registry contracts, not in glue UI**: AgentPass's SDK, Router402's facilitator fork, MCPay's proxy adapter, Soulmates' factory pattern, and ETL Network's threshold-signing pattern are the artifacts that compound. Third, **the Algorand and Bitcoin sides of mindX must be sourced largely off-platform**: ETHGlobal yielded only algobridge and AlgoSwap on Algorand and StarBTC + TitanFi + ThunderPortal on Bitcoin — for AlgoIDNFT and the deeper BTC primitives (BitVM, sBTC, Babylon, Citrea), Algorand Foundation hackathons and Bitcoin-L2 native ecosystems are the right hunting grounds. The strategic move for mindX is to **fork the Tier-S patterns into a unified reference architecture, port the x402 facilitator and HTLC patterns to Algorand/PyTeal, and use BONAFIDE's Latin-named contracts as the schema-stable wrapper around the borrowed primitives** — turning an inventory of hackathon code into a coherent autonomous-agent infrastructure within a quarter.